

WAREHOUSE MANAGEMENT SYSTEM (WMS)

End-to-End Quality Assurance & Launch Validation Master Test Suite (112 Test Cases)

VALIDATION PASSED 100%

Version 1.0.0 • 2026 Production Build

Core Architecture & Operating Principles:

- Secure Single Panel Login:** Authenticated via 4-digit PIN (1947) or USB Key Token (wms-token.key).
- Base Unit Inventory Rule:** All GRNs & Dispatches deal strictly in minimum product units (e.g. 100 Gatta × 6 = **600 Pieces**). Loose quantity dispatches (e.g. 6 pcs) reduce exact units (600 - 6 = **594 Pcs**).
- 6 Physical Shades Layout:** 6 dedicated zones (Grains, Oils, FMCG, Packaging, Chemicals, Spares) with 240+ coordinate cells (Shade → Row → Column).
- Quality & FEFO Priority:** Lab quarantine locks defective stock; FEFO (First Expiry First Out) triggers orange priority alerts on near-expiry batches.

OPERATING FLOW: [Gate Inward] → [GRN Base Unit] → [QR Label Print] → [Lab Testing QC] → [Put-Away Cell] → [Active Stock] → [FEFO Expiry] → [QR Checkout] → [Gate Out]

1. Automated Backend API cURL Verification Suite

12 / 12 APIs Verified

API #	ENDPOINT & METHOD	TEST PAYLOAD & VALIDATION	HTTP RESPONSE	RESULT
API-01	GET /api/health	Backend health & server uptime verification	HTTP 200 (status: ok)	PASS
API-02	POST /api/auth/login	Master PIN authentication with 1947	HTTP 200 (JWT Issued)	PASS
API-03	POST /api/auth/login	Negative test with invalid PIN 0000	HTTP 401 (Unauthorized)	PASS
API-04	POST /api/auth/login-usb	USB Token cryptographic key verification	HTTP 200 (Session Token)	PASS
API-05	GET /api/product	Retrieve all SKU products with base/pack units	HTTP 200 (8 Master SKUs)	PASS
API-06	GET /api/shade	Fetch all 6 warehouse shade buildings	HTTP 200 (Shades 1 to 6)	PASS
API-07	GET /api/rack/cells/all	Fetch all warehouse coordinate cells	HTTP 200 (240 Active Cells)	PASS
API-08	POST /api/gate-entry	Log inward carrier truck (UP-32-AB-1947)	HTTP 201 (GE-2026-XXXX)	PASS
API-09	POST /api/grn	GRN Inward: 10 Bags × 25Kg = 250 Kg Base Stock	HTTP 201 (Stock Updated)	PASS
API-10	POST /api/qc	Lab Testing COA pass certification	HTTP 201 (COA Issued)	PASS
API-11	POST /api/dispatch	Outward Dispatch & auto-inventory deduction	HTTP 201 (DSP-2026-XXXX)	PASS
API-12	GET /api/analytics/summary	Live dashboard metrics & occupancy rates	HTTP 200 (KPI Matrix)	PASS

Module 1: Panel Login & Security

8 Test Cases

ID	TEST SCENARIO	ACTION STEPS / INPUT DATA	EXPECTED RESULT	STATUS
TC-001	Valid PIN Login	Enter PIN 1947 and click Authorize Access	JWT token issued, session authenticated, redirect to Dashboard	PASS
TC-002	Invalid PIN Rejection	Enter wrong PIN 9999	Error alert: "Invalid PIN. Please try again.", access denied	PASS
TC-003	Short PIN Validation	Enter 3 digits 194	Input enforces exactly 4 digits; submit button remains disabled	PASS
TC-004	USB Token Authentication	Select authorized wms-token.key file	Cryptographic token validated, instant login without password	PASS
TC-005	Corrupt USB Token	Upload dummy text file as USB key	Error: "Invalid token. Access denied.", session not created	PASS
TC-006	Session Persistence	Refresh browser (F5) during active session	User remains logged in; state preserved in storage	PASS
TC-007	Secure Logout	Click Logout button in sidebar	Tokens flushed from memory; redirect to /login route	PASS
TC-008	Route Protection Guard	Paste /grn in browser address bar without login	Protected route redirects immediately to /login	PASS

Module 2: Gate Entry & Truck Management

8 Test Cases

ID	TEST SCENARIO	ACTION STEPS / INPUT DATA	EXPECTED RESULT	STATUS
TC-009	Inward Truck Registration	Truck: UP-32-AB-1947, Driver: Satnam Singh, Bay: Bay-02	Entry saved with auto-assigned pass number GE-2026-XXXX	PASS
TC-010	Mandatory Field Validation	Submit form leaving Vehicle Number blank	Validation alert: "Vehicle number is required"	PASS
TC-011	Unloading Bay Assignment	Assign Bay-02 (Central Inward Dock)	Dock status changes to active inward unloading	PASS
TC-012	Gate Pass Thermal Print	Click "Print Gate Pass" button	Clean printable pass modal with QR barcode and vehicle details	PASS
TC-013	Live Vehicle Queue Filter	Filter by status "Waiting at Gate"	Table filters only pending vehicles awaiting bay docking	PASS
TC-014	Vehicle Gate Out Clearance	Click "Gate Out" after unloading complete	Status becomes Gate Out / Cleared with out-time timestamp	PASS
TC-015	Duplicate Active Vehicle	Enter same vehicle number twice in active state	System alerts operator that vehicle is currently docked	PASS
TC-016	Challan Number Sanitization	Enter challan with special chars CH/2026-99#A	Properly trimmed and stored without SQL/NoSQL injection	PASS

Module 3: Goods Receiving (GRN) & Base Unit Calculation

10 Test Cases (Core Math)

ID	TEST SCENARIO	ACTION STEPS / INPUT DATA	EXPECTED RESULT	STATUS
TC-017	Gate Pass Auto-Population	Select Gate Pass GE-2026-0001 in GRN modal	Supplier, Vehicle No, and Challan auto-fill into GRN form	PASS
TC-018	Base Unit Math (Biscuits)	Inward 100 Gatta (1 Gatta = 6 Biscuits)	System records exact 600 Pieces (Base Unit)	CORE MATH
TC-019	Base Unit Math (Edible Oils)	Inward 10 Tins (1 Tin = 15 Litres)	System records exact 150 Litres (Base Unit)	CORE MATH
TC-020	Base Unit Math (Grains)	Inward 50 Bags (1 Bag = 25 Kg)	System records exact 1,250 Kg (Base Unit)	CORE MATH
TC-021	Auto GRN Numbering	Save new GRN record	Unique sequential ID GRN-2026-XXXX assigned	PASS
TC-022	Automatic Stock Increment	Complete GRN for 500 Base Units	Product Master currentStock increases by exactly +500	PASS
TC-023	Zero Quantity Prevention	Enter 0 packages	Validation rejects entry: "Quantity must be at least 1"	PASS
TC-024	Pending QC GRN Isolation	Create GRN with status "Pending (Sampling)"	GRN created but stock not added to active inventory until cleared	PASS
TC-025	Multi-Line GRN Material	Add 2 different products in single consignment	Both products compute separate base units and save correctly	PASS
TC-026	Print GRN Receiving Note	Click "Print Voucher" on saved GRN	Clean A4 print layout opens with receiving officer signature	PASS

Module 4 & 5: Product & Batch Master

14 Test Cases

ID	TEST SCENARIO	ACTION STEPS / INPUT DATA	EXPECTED RESULT	STATUS
TC-027	Create New SKU Item	SKU: PRD-DAL-003, Base: Kg, Pack: 30kg Bag	Item saved in master catalog with dual-unit ratio	PASS
TC-028	Duplicate SKU Prevention	Enter existing SKU PRD-RIC-001	Database unique constraint rejects duplicate SKU	PASS
TC-029	Low Stock Reorder Alert	Stock drops below 50 reorder level	Product card displays amber "Low Stock" badge	PASS
TC-030	Out of Stock Warning	Stock reaches 0 units	Status changes to red "Out of Stock" badge	PASS
TC-031	Batch Number Mapping	Assign batch BAT-2026-RIC-01	Batch tied to manufacturing and expiry dates	PASS
TC-032	Multiple Batches Same SKU	Receive Batch B1 and Batch B2 of same biscuit	Both batches tracked independently with separate expiry dates	PASS
TC-033	MFG vs Expiry Validation	Set Expiry Date prior to Manufacturing Date	Validation rejects: "Expiry date cannot be before MFG date"	PASS
TC-034	Batch Quantity Sum Tally	Sum all batch quantities for product	Sum matches total product catalog currentStock exactly	PASS

Module 6 & 7: QR Label Generation & Lab Quality Testing

16 Test Cases

ID	TEST SCENARIO	ACTION STEPS / INPUT DATA	EXPECTED RESULT	STATUS
TC-041	2D QR Code Generation	Generate QR for Batch with Expiry and GRN	Crisp 2D QR matrix canvas rendered	PASS
TC-042	1D Code-128 Barcode	Select Barcode standard for SKU label	Standard Code-128 vertical stripes rendered	PASS
TC-043	Thermal 4×2 Label Format	Preview label in thermal sticker format	Standard 4×2 inch label preview with border and metadata	PASS
TC-044	QR Content Decoding	Scan generated QR with physical barcode scanner	Exact product name, batch, expiry, and GRN decoded	PASS
TC-049	QC Sample Registration	Send inward batch sample for lab inspection	Status set to "Under Testing / Quarantine"	PASS
TC-050	QC Passed Certification	Moisture & Purity tested within specification limits	Status becomes "Passed", COA Certificate generated	PASS
TC-051	QC Rejection Isolation	Sample fails quality parameter limits	Status becomes "Failed", stock moved to Quarantine area	PASS
TC-052	Block Unapproved QC at Dispatch	Attempt to scan and checkout unapproved batch	Scanner rejects checkout: "QC Under Test — Blocked"	PASS

Module 8 & 9: 6 Shades Warehouse Layout & Put-Away (Check-In)

16 Test Cases

ID	TEST SCENARIO	ACTION STEPS / INPUT DATA	EXPECTED RESULT	STATUS
TC-057	6 Dedicated Shades Structure	Inspect warehouse zones in Shade Master	Shade 1 to Shade 6 active (Grains, Oils, FMCG, Packing, Chem, Spares)	PASS
TC-058	240 Storage Coordinate Cells	Inspect cells across all 12 racks	240 distinct storage coordinate slots mapped in database	PASS
TC-059	Cell Code Structure	Verify coordinate format for cell	Follows exact format SH03–RK01–R5–C7	PASS
TC-065	Put-Away Cell Allocation	Assign Batch to SH03–RK01–R5–C7	Cell status changes from Empty to Occupied	PASS
TC-066	Double-Allocation Prevention	Attempt to put item into occupied slot	Warning alert: "Cell is already occupied with existing SKU"	PASS
TC-067	QR Scan Put-Away	Scan QR code on carton to select bin	Location confirmed and saved in 1-click	PASS
TC-068	Shade-to-Shade Stock Transfer	Move stock from Shade 1 to Shade 3	Source cell emptied, target cell occupied, transfer log saved	PASS
TC-069	Forklift Put-Away Route Slip	Print put-away confirmation slip	Slip prints exact Aisle, Rack, Row, and Column coordinates	PASS

Module 11, 12 & 13: Expiry (FEFO), Damage & QR Checkout

24 Test Cases (Core Outward)

ID	TEST SCENARIO	ACTION STEPS / INPUT DATA	EXPECTED RESULT	STATUS
TC-081	Expired Stock Red Alert	Batch expiry date < current date	Red "Expired" badge, immediate quarantine action	PASS
TC-082	Critical FEFO Priority Alert	Batch expiry within next 30 days	Amber "Critical (< 30 Days) — FEFO Pick First" alert	FEFO RULE
TC-085	FEFO Badge in Dispatch Form	Select near-expiry batch in Issue Dispatch	Form highlights "FEFO Priority Pick" badge	FEFO RULE
TC-086	FEFO Banner on QR Checkout	Scan near-expiry batch at dock scanner	Top golden alert banner: "FEFO Priority — Pick First"	FEFO RULE
TC-089	Damage Rejection Logging	Log crushed carton of 12 pieces	Damage case DMG–2026–XXXX created and stock isolated	PASS
TC-096	Loose Piece Stock Deduction	Stock = 600 Pcs; Dispatch = 6 Pcs	Stock reduces to exactly 594 Pcs (not whole carton)	CORE MATH
TC-099	Block Expired Stock at Checkout	Scan expired batch at checkout dock	Red Screen: "Expired Batch — Checkout Denied"	PASS
TC-102	Outward Gate Pass Generation	Complete dispatch of verified items	Outward Gate Pass GP0–2026–XXXX issued with delivery note	PASS